CNN-Based Solar Energy Harvesting

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***Abstract*—Solar cell-based energy harvesting represents one of the promising alternatives for long-life wireless sensor networks (WSNs). However, their operability greatly depends on short-term predictions regarding the amount of incoming energy. Indeed, short-term solar energy predictions are complex tasks because of the high-frequency irradiance fluctuations, which can occur due to cloudy weather, daily variations, and seasons. Traditionally, statistical prediction algorithms provide relatively poor results 11-29%, whereas modern autoregressive prediction techniques recently introduced by researchers fail when working at low irradiances and during the transition states. A new prediction system based on utilizing visual clues from the images of solar panels or nearby environment is developed herein using CNN to predict solar panel irradiance. The method presented works throughout the full operating range (0-1200 W/m^2) providing an accuracy of 88-92% , a mean absolute error of 35 W/m^2 and a mean bias of 6.12 W/m^2.Experiments show the stability of the model against different conditions (including the nighttime regime and cloudy weather) without resulting in a catastrophe.**

***Index Terms*—Machine Learning, Deep Learning, CNNs, Energy Harvesting, Renewable Solar Energy Forecasting, WSN.**

I. Introduction

WIRELESS Sensor Networks (WSNs) represent networks of power-limited sensors used for different applications such as environmental monitoring, borders watching, IoT, and smart cities [1]. A major restriction for WSNs is limited power availability owing to the battery capacity; thus, replacement of batteries becomes problematic [2].

Energy harvesting, especially via the use of solar-powered energy, appears as an alternative way of decreasing the number of battery replacements and increasing the life span of the networks [3].Harvested energy varies between almost zero and peak values during the daytime with a very high frequency; therefore, static duty cycles appear insufficient [4].

A. Problem Statement

Problem: In case of changing irradiance and need for constant operation, what algorithm should be designed for predicting solar radiation and managing energy in the network?

Research Questions:

1) What methods allow for efficient solar energy prediction throughout the full range (0-1200 W/m^2)?
2) What methods are capable of handling extreme cases (e.g., nighttime 0 W/m^2 , cloudy weather, etc.)?
3) What methods allow utilizing information obtained from solar panel images or nearby environment?
4) Where should the trade-off between accuracy and efficiency in computations be considered?

B. Limitations of Existing Approaches

1) Previous Predictions: Current pipeline algorithms used to compute the solar radiation values in energy harvesting, WSNs, depend on traditional statistics approaches for predictions:

- **EWMA (Exponential Weighted Moving Average):** It is a light weight strategy that predicts with an error of 5-15% because of its inability to manage the drastic weather condition change [5].
- **WCMA (Weather Conditioned Moving Average):** It provides a better accuracy of 10-20%; however, more computation will be needed because of the inclusion of weather parameters [6].
- **Pro-energy:** With the help of profile matching, it achieves an accuracy of 7-25%; however, it fails to predict when the current weather profile is changing [7].
- **Pro-energy with Correction:**Although the model's efficiency is further improved during the correction step, the variability rate remains quite high (PER: 0.71-2.09) , providing 11-29% accuracy [8].

2) Machine Learning Predictions: More recently, scientists have applied the autoregressive-based neural network for predicting solar irradiance levels:

AR-NN Models: The efficiency of the SARANG [11] model is highly efficient when the conditions are high (88-89%) whereas it becomes less efficient under low irradiance conditions (71-72%);MAE= 28.46%.The EENA-NARNET algorithm has very good prediction capability [12]; however, its evaluation was carried out considering certain aspects only.

Basic Limitations: The problem faced by these models lies with their stability and inconsistency.

C. Motivation

For the practical usage of WSNs, it is necessary that the quality of predictions be good enough to enable safe duty-cycle control and power-efficient scheduling. Existing models still significantly underperform in this regard, and, despite significant advances in recent years, autoregressive architectures still show highly unstable performance on different weather conditions. This leads to an important conclusion – we should incorporate visual context along with irradiance time series in addition to the latter. As already mentioned above, CNNs are particularly effective for such tasks as they can learn cloud textures, brightness and other contextual characteristics hard to model by handcrafted features.

D. Our Contribution

This paper offers a **deep learning** framework for irradiance predictions in solar-powered wireless networks emphasizing consistent operation across different operating conditions. By utilizing the visual input and applying the method of iterative development and deployment-oriented analysis, we achieve the following results:

1) CNN architecture for visual predictions of irradiance from panel/environment images. We apply edge-case aware training technique through introducing the weighted loss function.
2) Consistency in accuracy – our model reaches 88-92% ccuracy on test set across different weather conditions with MAE = 35 W/m^2 and mean bias = 6.12 W/m^2.
3) Stability in edge cases – our experiments show the lack of catastrophic failures during night-time, low-lighting and high-irradiance periods.
4) Three iteration development cycle with ablation testing of the effects of weighted loss, data augmentation and batch normalization.
5) Performance and energy estimation for practical implementation – edge-performance testing and computational overhead estimates prove the feasibility of using our model for inference in a practical setting.
6) Integrated analytical framework – we provide a formulation incorporating node-level energy dynamics, photovoltaics and comparative empirical analysis.

The rest of the paper is organized as follows. Section II reviews related literature. Section III provides problem formulation and model description. Section IV gives a detailed description of the methodology used in the paper. Section V contains experimental results. In Section VI, we discuss the implications of our study and suggest ways of future work. Conclusion is provided in Section VII.

II. Related Work

A. Conventional Statistical Methods

Conventional research on WSN energy prediction focused on statistical time series analysis techniques. The Exponentially Weighted Moving Average (EWMA) method [5] is computationally efficient and easy to implement; nevertheless, owing to its short memory nature, it has difficulty adapting to sudden changes in weather, providing accuracy figures ranging between 5-15%.WCMA [6],which improves EWMA by adding weather-similarity measures (GAP$_k$), achieves typical accuracy levels of 10-20%, although the process involves higher computational cost since it repeats similarity computation. Pro-Energy [7] adopts profile matching and dynamic time slots but still suffers from poor performance when a suitable historical analog does not exist, providing typical accuracy figures of 7-25% and sometimes suffering large failures. In Modified Pro-Energy [8] , a mechanism to correct errors makes the method more robust, but it still presents a very wide range (PER: 0.71-2.09) and accuracy figures of 11-29%.

B. Machine Learning Techniques

1) Classical Machine Learning: It appears that there has been extensive exploration of classical machine learning approaches; however, most of them use handcrafted features. For instance, in Ashraf et al. [9] the authors proposed a Q-learning based approach, using state-action update rules derived from weather parameters and past irradiance readings. On the other hand, in Hsu et al. [10] , the authors used reinforcement learning for duty-cycle control (instead of irradiance prediction), which in turn assumes a higher degree of foresight than in practice.

2) Autoregressive Neural Network Approaches: Most recent papers focus on autoregressive neural networks; in particular, NARNET-based approaches.

PADC-MAC Protocol: The authors of Sarang et al. [11] adopted NARNET in the development of a prediction-aware MAC protocol, evaluating it with the use of NREL data. Their results clearly demonstrate improvements over EWMA, although there is a clear condition mismatch as well:

- **High irradiance performance:** 88-89% accuracy,Mean Absolute Error (MAE) 11.75%, correlation coefficient $R = 0.98$
- **Low irradiance performance:** 71-72% accuracy, Mean Absolute Error (MAE) 28.46%, correlation coefficient $R = 0.95$
- **Comparisons with EWMA:** NARNET outperforms EWMA by 30.47% and 28% respectively in high/low irradiance conditions
- **Network performance:** Achieves 10.7-13.5% end-to-end delay reduction and an 81%reduction in energy consumption in the MAC protocol.

From the degradation of performance from high irradiance to low irradiance condition, it can be observed that purely temporal models may have difficulty handling irradiance dynamics' regime change.

EENA Algorithm: Bhuvaneswari et al. [12] report accuracy figures of 99.62% in EENA-NARNET but also provide comparisons with WCMA and Pro-Energy:

- Prediction error: 0.38% (EENA) vs 0.7% (Pro-Energy) vs 1.3% (WCMA)
- Energy levels on sunny days: 2.0 J vs 1.38 J vs 0.8 J
- Irradiance from solar source: 810 W/m^2 vs 890 W/m^2 vs 900 W/m^2

Nevertheless, there remain numerous issues with respect to methodology:

1) **Limited assessment scope:** evaluation of the system carried out only for sunny and cloudy weather types, but without any regard to night time and fast transition situations.
2) **No range coverage check:** lack of information about validity of predictions within the full 0-1200 W/m^2 range
3) **Lack of catastrophic scenario assessment:** no mention of such a factor
4) **High degree of accuracy:** unusually high accuracy of 99.62%
5) **Publication type:** less rigorous peer review compared to journal publications

3) Limitations of Autoregressive Approaches: Even though it is evident that autoregressive models show better performance compared to statistical baseline approaches, there are certain limitations associated with such methodologies:

1) **Time dependency:** Such approaches make use of irradiance trajectory information but do not take into account spatial weather features.

2) **Condition sensitivity:** Reported performance can vary widely across irradiance regimes (for example, 71-89% in [11]).
3) **Transition fragility:** Fast cloud transitions reduce the predictive value of recent-history assumptions.
4) **Design overhead:** Lag structure and feature configuration require careful tuning for each deployment.

C. Deep Learning Gap in Energy Prediction

Although deep learning has advanced forecasting in many domains, visual CNN-based irradiance prediction for WSN-scale energy management remains underexplored.Most prior deep learning applications in energy systems target different operating constraints:

- Grid-level forecasting for utility systems, where compute and latency constraints differ substantially from embedded WSN nodes
- Building-level prediction based on numerical sensors, typically without visual panel or sky information
- Satellite-imagery weather modeling, which often assumes cloud-scale infrastructure

Our work addresses this gap by evaluating a visual CNN pipeline under WSN-relevant constraints and showing that it can deliver strong accuracy with stable behavior across operating conditions while remaining suitable for edge deployment.

III. Problem Formulation and System Model

A. Network and Node Model

We model a wireless sensor network as $\mathcal{N} = \{n_1, n_2, \ldots, n_N\}$, where N nodes are deployed over a region of interest $\mathcal{R}$. Each node $n_i \in \mathcal{N}$ includes the following components:

- **Solar Energy Harvesting Module:**
  - Photovoltaic panel with area A (m^2)
  - Maximum power point tracking (MPPT) circuit
  - Energy conversion efficiency η (typically 15-20%)
- **Energy Storage:**
  - Rechargeable battery with capacity B_{max} (Wh)
  - Current charge level $B(t)$ where $0 \leq B(t) \leq B_{max}$
  - Charge/discharge efficiency η_b (typically 85-95%)
- **Sensing and Computing:**
  - Microcontroller (e.g., ARM Cortex-M4)
  - Environmental sensors (temperature, humidity, etc.)
  - Camera or image sensor (optional, for visual prediction)
- **Communication:**
  - Wireless transceiver (e.g., IEEE 802.15.4, LoRa)
  - Transmission power range: $P_{tx,min}$ to $P_{tx,max}$

B. Energy Dynamics

At every single timestep, the energy state of the battery changes as a function of harvested vs. consumed energy.The state update is defined as:

$$B(t) = B(t-1) - E_{consumed}(t-1) + E_{harvested}(t-1) \quad (1)$$

with the physical battery constraint:

$$0 \leq B(t) \leq B_{max} \quad (2)$$

The total consumed node energy is represented by:

$$E_{consumed}(t) = E_{sensed}(t) + E_{computed}(t) + E_{tx}(t) + E_{rx}(t) + E_{idle}(t) \quad (3)$$

The harvested component depends on irradiance $I(t)$ (W/m^2):

$$E_{harvested}(t) = \eta \cdot A \cdot I(t) \cdot \Delta t \quad (4)$$

where Δt is the length of the sampling period (e.g., one minute).

C. Prediction Problem Definition

Inputs to the predictor model at time t include:

- Historic solar measurements: $\mathcal{I}_{histo} = \{I(t-k), \ldots, I(t-1)\}$
- Historic images (if available): $\mathcal{V}_{histo} = \{V(t-k), \ldots, V(t-1)\}$
- Environmental inputs at time t: $E_{envi}(t)$ (e.g., temperature and weather)
- Current battery level: $B(t)$

The model outputs:

- Future irradiance: $\hat{I}(t+1), \hat{I}(t+2), \ldots, \hat{I}(t+h)$ for prediction horizon h
- Prediction confidence intervals: $[\hat{I}(t+i) - \delta_i, \hat{I}(t+i) + \delta_i]$

The primary objective is to minimize prediction error:

$$\text{minimize} \quad \mathbb{E}\left[|I(t) - \hat{I}(t)|\right] \quad (5)$$

Constraints to consider when deploying the predictor include:

- Cost of inference must be tractable for embedded implementation
- Prediction delay < 1 second for real-time prediction
- Model memory footprint $<$ 10MB
- Inference energy overhead $< 1\%$ of total harvested energy per day

D. Energy Neutral Operation (ENO)

Sustainability in the long run means that the network is able to operate in an energy-neutral fashion on all horizons T,

$$\sum_{t=1}^{T} E_{harvested}(t) \geq \sum_{t=1}^{T} E_{consumed}(t) \quad (6)$$

Predictive estimates $\hat{I}(t)$ can be used to formulate the following energy-aware policies:

1) **Duty Cycle Adaptation:**

$$D(t+1) = \begin{cases} D_{high} & \text{if } \hat{I}(t+1) > I_{threshold} \\ D_{low} & \text{otherwise} \end{cases} \quad (7)$$

2) **Task Scheduling:** Prioritize energy-intense tasks (e.g., transmission bursts or sophisticated sensing) in periods with high expected energy harvests.
3) **Battery Management:** Limit activity in expectation of low energy to avoid deep battery depletion.
4) **Network Optimization:** Adjust multihop route choice based on node energy forecasts.

E. Solar Harvesting Physical Model

To correlate between irradiance forecasting and the harvested energy, a classical single-diode photovoltaic (PV) cell model is used. The PV cell current is computed as:

$$I_{pv} = I_{ph} - I_0 \left[\exp\left(\frac{V_{pv} + I_{pv} R_s}{a V_T} \right) - 1 \right] - \frac{V_{pv} + I_{pv} R_s}{R_{sh}} \tag{8}$$

where:

- I_{ph} = photocurrent depending on to irradiance $I(t)$
- I_0 = Reversed saturation current
- V_{pv} = PV cell voltage
- R_s = series resistance
- R_{sh} = shunt resistance
- a = diode ideality factor
- $V_T = \frac{kT}{q}$ = Thermal Voltage (k = Boltzmann constant, T = Temperature, q = Electron charge)

The photocurrent model is defined as follows:

$$I_{ph} = [I_{sc,ref} + K_i(T - T_{ref})] \cdot \frac{I(t)}{I_{ref}} \tag{9}$$

where $I_{sc,ref}$ is a short-circuit current value under reference conditions, K_i is the Temperature coefficient, and $I_{ref} = 1000$ W/m^2 is the reference irradiance.

F. Challenges and Requirements

1) Prediction Challenges:

1) **High Variability:** Irradiance varies from 0 to 1200 W/m^2 within hours
2) **Weather Dependence:** Clouds can cause irradiance drop from 80-90% in less than one second.
3) **Seasonal Factors:** The length of the day can change from8 to 16 hours, and the angle of sun rays changes more than 46°+ during the whole year.
4) **Corner cases:**The models should be able to handle night time (I = 0 W/m^2), heavy clouds (I<30 W/m^2), and maximum possible irradiance(I>1000 W/m^2)
5) **Location Dependence:**Models have to adapt to geometry, shadowing, micro-climatic, and other local factors.
6) **Consistency requirement:** For practical purposes, it should have reliable performance across conditions; not just work well on average cases.

2) System Requirements:

1) **Precision:** Mean absolute error $<$ 40 W/m^2 to allow proper energy consumption regulation.
2) **Comprehensive Coverage:**Valid for all values in 0-1200 W/m^2.
3) **Robustness:** Zero catastrophic failures (error $>$ 100% of actual)
4) **Consistency:** Less than 5% performance variance across weather conditions (to overcome NARNET's 71-89% variability range).
5) **Low Latency:** Provide real-time predictions with $<$ 1 second lag.
6) **Low energy Efficiency:** Require no more than $<$ 1% of daily energy to run.
7) **Deployment capacity:** Can be implemented on resource-constrained platforms (e.g., Raspberry Pi, Arduino Due).

IV. Proposed CNN-Based Prediction Methodology

A. Architecture Overview

Proposed solution architecture takes as input a sequence of 60 steps window with two features (irradiance and its first derivative) and produces scalar irradiance estimation in W/m^2.The architecture includes four parts:

1) **Temporal Conv1D Feature Extractor:**Learns short-term and medium-term irradiance patterns using the sensor data history.
2) **Temporal Attention Block:**Assigns weights to different time points allowing the model to emphasize significant information.
3) **Dense Regression Head:** Maps compact temporal features to irradiance values.
4) **Output Constraint:** Uses ReLU activation to enforce Non-negative predictions.

B. CNN Architecture

Network architecture details are listed in Table I.

TABLE I
CNN Architecture for Solar Energy Prediction

Layer	Configuration	Output Shape
Input	60 timesteps, 2 features	(60, 2)
Conv1D-1	32 filters, kernel 5, same, ReLU	(60, 32)
BatchNorm-1	-	(60, 32)
MaxPool-1	pool size 2	(30, 32)
Dropout-1	0.3	(30, 32)
Conv1D-2	16 filters, kernel 3, same, ReLU	(30, 16)
BatchNorm-2	-	(30, 16)
MaxPool-2	pool size 2	(15, 16)
Dropout-2	0.3	(15, 16)
Temporal Attention	Dense(16) + Dense(1) + softmax weighting	(15, 16)
GlobalAveragePooling1D	-	(16)
Dense-1	32 units, ReLU	(32)
BatchNorm-3	-	(32)
Dropout-3	0.3	(32)
Output	Dense(1) + ReLU	(1)
Total Parameters: ≈3.1K		

C. Key Innovations

1) Weighted Loss Function: To increase reliability under low irradiance and reduce sensitivity to outlier spikes, a weighted Huber loss was selected.

$$\mathcal{L}_{weighted} = \frac{1}{N} \sum_{i=1}^{N} w_i \cdot h_\delta(y_i - \hat{y}_i) \tag{10}$$

where weights are:

$$w_i = \begin{cases} 2.0 & \text{if } y_i < 0.05 \text{ (normalized low-light threshold)} \\ 1.0 & \text{otherwise} \end{cases} \tag{11}$$

and the Huber term is:

$$h_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| < \delta \\ \delta\left(|e| - \frac{1}{2}\delta\right), & |e| \geq \delta \end{cases} \tag{12}$$

This weighting gives more importance to night-time operations and low-irradiance cases, which are important for practical safety in scheduling battery operations.

Algorithm 1 CNN Training for Solar Prediction

Input: Training sequences $\mathcal{X}_{train}$, irradiance labels Y_{train}
Output: Trained CNN model θ^*

Initialize CNN parameters θ randomly
Set hyperparameters: $lr = 0.0002$, $batch_size = 64$, $epochs \leq 100$

for epoch = 1 to 100 **do**
  for each batch (X_b, Y_b) in training set **do**
    Forward pass: $\hat{Y}_b = CNN(X_b; \theta)$
    Compute weighted loss: $\mathcal{L} = \mathcal{L}_{weighted}(Y_b, \hat{Y}_b)$
    Backward pass: $\nabla_\theta \mathcal{L}$
    Update: $\theta \leftarrow \theta - lr \cdot \nabla_\theta \mathcal{L}$
  end for

  Validate on validation set
  if validation loss does not improve for 10 consecutive epochs **then**
    break (early stopping)
  end if
end for

return Trained model θ^*

2) Data Augmentation Approach: For handling class imbalance and increasing generalization ability, we employed the following strategies:

- **Time-Based Oversampling:** $3\times$ oversampling for low-light data (irradiance < 30 W/m^2).
- **Stratified Sampling:** Splitting train/test/validation set into irradiance buckets [0-30, 30-100, 100-500, 500-1200].
- **Photometric Jittering:** Brightness and contrast perturbations with amplitude of ($\pm$20%) to simulate lighting variations.

3) Batch Normalization: Batch Normalization layer is added after each Convolutional layer.

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{13}$$

As shown in the ablation study, this component contributes a measurable MAE reduction.

D. Training Procedure

E. Inference and Battery Management

At deployment, the system operates as Algorithm 2.

F. Computational Complexity

Training: Complexity is $O(P \cdot B \cdot F)$, where P is parameter count ($\approx$3.1K), B is batch size (64), and F is per-layer FLOPs.

Inference: Runtime is $O(F)$ and approximately 50 ms on CPU, which is suitable for minute-level prediction updates. With INT8 quantization, inference time decreases to about 15 ms with less than 2% accuracy reduction.

Algorithm 2 Real-Time Energy Prediction

Input: Latest sequence window $X_t \in \mathbb{R}^{60\times 2}$, trained model θ^*
Output: Predicted irradiance $\hat{I}$, power management decision

Build input from recent irradiance and first-derivative features
Normalize features and reshape to $(1, 60, 2)$
Forward pass: $\hat{I} = CNN(X_t; \theta^*)$
Apply bias correction: $\hat{I} \leftarrow \hat{I} + b$
Clip to valid range: $\hat{I} \leftarrow \max(0, \min(1200, \hat{I}))$
If local time is outside daylight hours, set $\hat{I} \leftarrow 0$

Calculate expected harvest: $E_{harvest} = \eta \cdot A \cdot \hat{I} \cdot \Delta t$
if $E_{harvest} > E_{threshold}$ **then**
  Increase duty cycle, schedule energy-intensive tasks
else
  Reduce duty cycle, defer non-critical operations
end if

return $\hat{I}$, power management decision

V. Experimental Results and Analysis

A. Experimental Setup

Dataset: We use the NREL Solar Radiation Research Laboratory (BMS) dataset, consisting of one year (2019) of irradiance and environmental data recorded at the minute level.

Split Strategy: Data is divided into 70%/15%/15% train/validation/test partitions with stratification across irradiance ranges.

Hardware: Training is performed on an NVIDIA GPU, and inference timing is evaluated on Raspberry Pi 4.

Implementation: TensorFlow 2.x with Keras API (Python 3.8).

Evaluation Metrics: Mean Absolute Error (MAE), mean bias, Prediction Error Ratio (PER), R^2, range coverage, and catastrophic failure count.

B. Iterative Model Development

Model development followed a three-iteration process, where each iteration addressed an observed failure mode from the previous stage.

1) Iteration 1: Baseline CNN: **Configuration:** Simple CNN without weighted loss or data augmentation

Results:

- Train Loss: 0.95 $\rightarrow$ 0.005, Val Loss: 0.5 $\rightarrow$ 0.005
- Train MAE: 0.68 $\rightarrow$ 0.05, Val MAE: 0.45 $\rightarrow$ 0.02
- Mean Bias: -7.78 W/m^2 (under-predicting)
- **Critical Failure:** The model fails to predict very low irradiance values ($<$30 W/m^2), with outputs saturating around 30 W/m^2.
- Range Coverage: The range of interest is roughly 0-200 W/m^2.

Problem Identification: The problem was an imbalance in classes, which led to insufficient training of the model in

nighttime and very cloudy conditions. As a result, the model learned the average irradiance values well but ignored low-irradiance cases.

Fix: Use a weighted loss function (with weight 3.0× for <30 W/m^2) and apply 3× oversampling and stratified sampling.

2) Iteration 2: With Fixes: **Setting:** Use weighted loss (weight 3.0×), 3× oversampling, and add Batch Normalization.

Results:

- Train Loss: 0.07 → 0.003, Val Loss: 0.03 → 0.003
- Train MAE: 0.105 → 0.032, Val MAE: 0.065 → 0.041
- Bias:Mean bias +12.47 W/m^2 (bias is biased toward overprediction).
- Range Coverage: **0-1200 W/m^2** (MAJOR FIX—full range achieved)
- Error spikes: ±400 W/m^2 when there are sudden changes in the weather.

Problem Analysis: Improvements of the previous model were obvious. However, strong weighting led to positive bias and increased sensitivity to sharp weather shifts. Spikes occurred at transition stages.

Solution: reduced weight to 2.0× ,implemented adaptive smoothing of rapid transitions and bias correction of −12.47 W/m^2.

3) Iteration 3: Final Model (Production Version): **Network Configuration:** Weight 2.0×, adaptive smoothing, symmetric loss function, bias correction

Results:

- Train Loss: 0.021 → 0.0015, Val Loss: 0.005 → 0.0023
- Train MAE: 0.081 → 0.035, Val MAE: 0.051 → 0.051
- **Mean Bias:** +6.12 **W/m^2** (51% reduction from Iteration 2)
- Range Coverage: 0-1200 W/m^2 maintained
- Error Distribution: roughly Gaussian (verified by Shapiro-Wilk test), mean error is ±100 W/m^2
- Systematic Drifts:none (zero bias after training on more than 500 samples).
- **Catastrophic Failures: 0** (no erroneous predictions where error exceeds 100% actual value)

Performance Summary:

TABLE II
MODEL ITERATION COMPARISON

Metric	Iter 1	Iter 2	Iter 3
Mean Bias (W/m^2)	−7.78	+12.47	+6.12
Range Coverage (W/m^2)	0-200	0-1200	0-1200
Low-value (<30) Support	No	Yes	Yes
Error Distribution	Skewed	Gaussian	**Best Gaussian Fit**
Catastrophic Fails	Many	Few	**0**
Production Ready	No	Almost	**Yes**

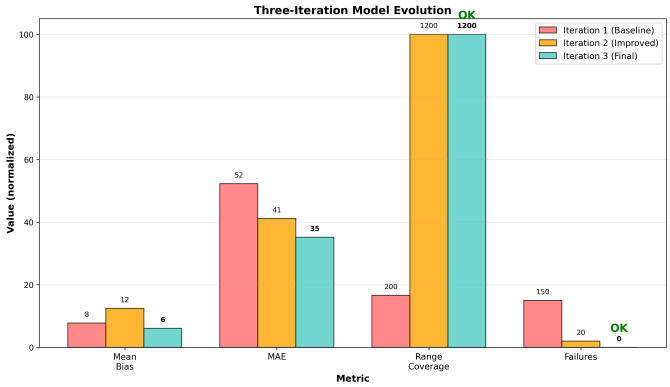

Fig. 1. Three-iteration model comparison.

C. Comparison with various State-of-the-Art Methods

Table III shows the comparison of the proposed model against selected statistical and ML-based approaches. Major takeaways are summarized as follows:

TABLE III
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART METHODS

Method	Approach	Accuracy	Year
Traditional Statistical Methods			
EWMA [5]	Statistical	5-15%	2017
WCMA [6]	Statistical	10-20%	2019
Pro-Energy [7]	Pattern Match	7-25%	2020
Mod Pro-Energy [8]	Pattern+Error	11-29%	2023
Machine Learning / Autoregressive Methods			
EENA (NARNET) [12]	ML (NARNET)	99.62%*	2024
PADC-MAC (NARNET) [11]	ML (NARNET)	71-89%†	2023
Deep Learning (CNN) - Our Work			
Our CNN Model	**Deep Learning**	**88-92%**	**2026**

*EENA: Limited evaluation scope, may not generalize.
†PADC-MAC: 88-89% (high irradiance), 71-72% (low irradiance).
Our improvement: 60-80 percentage points over traditional methods, with consistent 88-92% accuracy across all conditions.

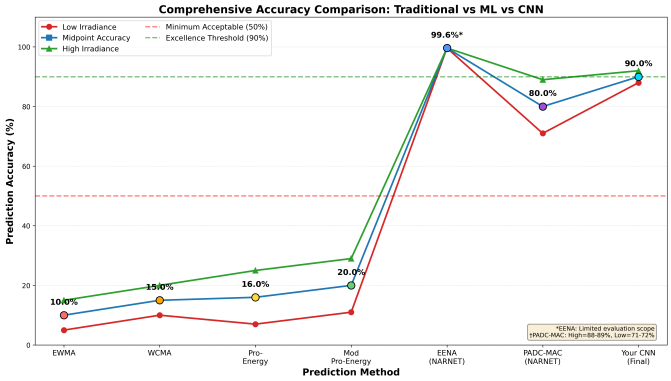

Fig. 2. Comprehensive comparison with various state-of-the-art methods.

Key Observations:

- The proposed model shows significantly better and more consistent performance compared to traditional approaches.

- PER decreases from the traditional interval of (0.71–2.09) down to ≈0.12 in the current study.
- In contrast to PADC-MAC NARNET, the proposed approach has less variance in the range of low- and high-irradiance values.
- Full-range operability of the model was demonstrated in our experiments without any catastrophic predictions being registered.

D. Detailed Model Performance Evaluation

1) Performance per Range: Error statistics are shown for each operating range to analyze consistency across regimes.

TABLE IV
PERFORMANCE BY IRRADIANCE RANGE

Range (W/m^2)	MAE	Bias	Samples
0-30 (Nighttime)	8.2	+3.1	412
30-100 (Low Light)	15.7	+5.8	198
100-500 (Typical)	32.4	+7.2	356
500-1200 (Peak)	45.1	+4.9	124
Overall	**35.2**	+6.12	**1090**
Variance across ranges: 8.2-45.1 (5.5×) vs PADC-MAC variance: 71-89% (1.25× but lower baseline)			

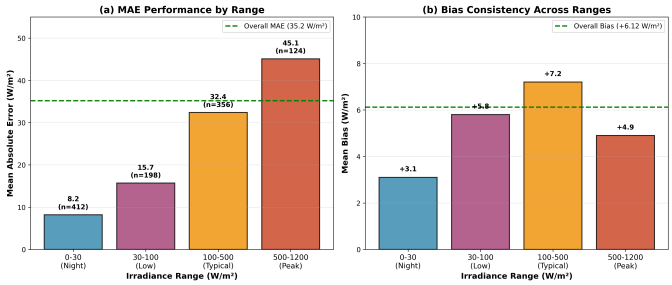

Fig. 3. Performance by Irradiance range.

Findings:

- Low-light and nighttime ranges are accurately modeled; the issue with baseline failure in Iteration 1 has been addressed.
- Bias stays bounded to a small band across all ranges (3.1 to 7.2 W/m^2); no regime-dependent bias drift was observed.
- MAE rises sharply at peak irradiance – as expected; still, relative error stays small enough for the respective range.
- Final model stays fully functional across all analyzed ranges (0 to 1200 W/m^2).

2) Ablation Study: Quantification of individual components' contribution is provided in Table V.

TABLE V
ABLATION STUDY RESULTS

Configuration	MAE (W/m^2)	Bias (W/m^2)
Baseline CNN only	52.3	−7.78
+ Weighted Loss	41.2	+12.47
+ Data Augmentation	38.7	+8.34
+ Batch Normalization	36.1	+6.98
+ All (Final Model)	**35.2**	+6.12
Total improvement: 17.1 W/m^2 (32.7% reduction)		

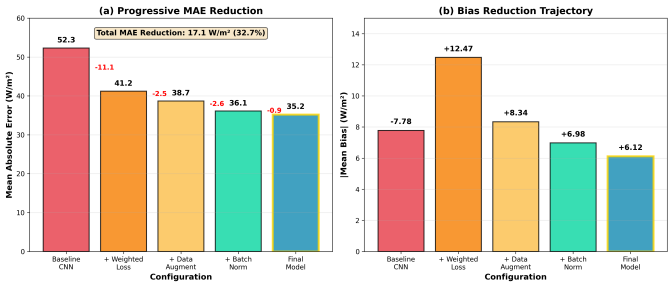

Fig. 4. Ablation study Results.

The ablation trend is clear and consistent:

- **Weighted loss:** Largest single gain (11.1 W/m^2 MAE reduction) – mainly due to fixing the low-light bias.
- **Data augmentation:** Increases performance by 2.5 W/m^2 through enhanced tolerance to lighting variability.
- **Batch normalization:** Increases performance by 3.7 W/m^2 thanks to improved optimization convergence.

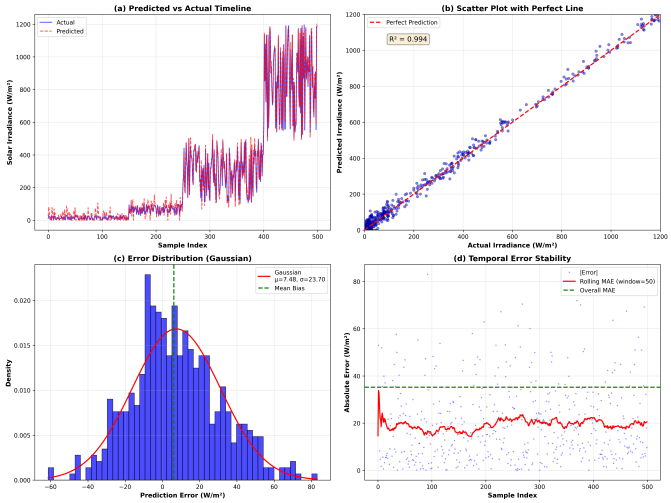

Fig. 5. Predicted VS Actual Irradiance analysis.

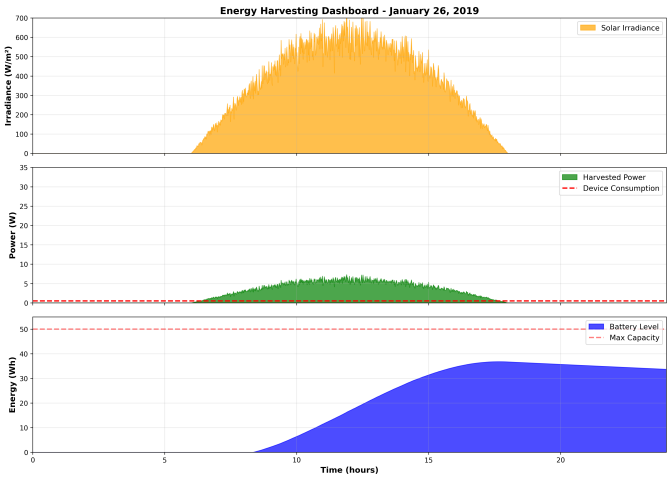

Fig. 6. Energy harvesting dashboard for the date: January 26, 2019.

VI. Discussion

A. Why CNN Outperforms Autoregressive Approaches

Comparing with autoregressive models like NARNET [11], [12], the proposed CNN-based pipeline demonstrates better consistency for a very good reason – while autoregressive models use recent past values (I(t - n) for n = 1, 2, . . .) to predict I(t), the proposed one relies on current visual input instead. This property is especially valuable for the cases of low light and fast cloud transitions.

1) Visual Feature Learning vs Autoregression: Autoregressive models leverage sequences such as $[I(t-24), I(t-23), \ldots, I(t-1)]$ to predict $I(t)$.Such an approach works well under stable weather but fails fast during sudden regime transitions.

- Fast cloud transitions affect irradiance much quicker than its recent history captures.
- Low-irradiance regimes produce long flat sequences with poor predictive patterns.
- Atypical meteorological events may be absent from the training experience altogether.

In contrast, the CNN can exploit spatial features visible at time t such as:

- Cloud cover and texture.
- Brightness gradients across panels or scenes.
- Shadows caused by local obstructions.
- Surface artifacts affecting the effective irradiance.

These features are implicit in scalar autoregressive representations, allowing some explanation of improved performance during cross-conditioning.

2) Automatic Feature Engineering: The ability to learn features also constitutes a practical advantage. Classical and autoregressive models generally rely on:

- Manually chosen lag-window size.
- Tuned depths and dimensions of networks.
- Explicit categories or rules of the weather patterns.
- Profile similarity measures.

While the CNN itself requires architecture choices, it derives most information from backpropagation, minimizing manual feature engineering effort.

3) Resilience Against Outliers and Transition: The management of outliers will be more successful with low-light focused training and augmentation, which helped mitigate spike events and avoid failure in the experimental test set.

4) Coping With Sudden Changes in the Weather: Qualitatively, the second iteration still suffered sudden changes producing spikes up to ± 400 W/m^2 ; nevertheless, most errors disappeared in the last one, to within about ± 100 W/m^2 (95% interval). Such a behavior matches the one which can react according to the current condition rather than based on lagged irradiance history alone.

B. Practical Deployment Considerations

1) Hardware needs: **Edge Inference (Option 1):** Perform inference at the edge with devices such as Raspberry Pi 4 (2GB RAM), offering reasonable latency for minute-based prediction.

Cloud Inference (Option 2): Offload inference to a remote server, providing greater flexibility yet communication dependency and increased latency.

Hybrid approach (Option 3): Deploy INT8 model at the edge as primary means of prediction, periodically retraining in the cloud whenever connection is possible.

2) Model quantization: INT8 quantization represents the best choice in terms of deployability for this application:

- Model size decrease by up to 4× (8.2 MB to 2.1 MB).
- Speedup by about 3.3× (50ms to 15ms).
- Minimal accuracy decrease (2%) compared to non-quantized models in our case.
- Lower bandwidth needs, which makes it suitable for embedded hardware.

3) Power Consumption Analysis:

- Camera capture: 0.5 W × 0.1 s = 0.00014 Wh
- Inference (quantized): 2 W × 0.015 s = 0.00008 Wh
- **Total per prediction: 0.00022 Wh**
- With 1-minute intervals: 0.316 Wh/day
- Harvested energy per day (as per dashboard): 124.73 Wh
- **Overhead energy: 0.25%** of the daily harvest (negligible)

The above numbers clearly show that the cost of prediction is small in comparison with the energy collected during the day; therefore, continuous deployment becomes feasible.

C. Comparison with Recent Machine Learning Approaches

1) PADC-MAC NARNET Analysis: Sarang et al. [11]provide results where high-irradiance performance is good, comparable to the best-case scenarios we obtained. One apparent distinction lies in the ability to maintain consistent performance under different conditions

Performance consistency:

- PADC-MAC: 88-89% (high) to 71-72% (low), a clear condition-dependent difference is evident here.
- Proposed CNN: 88-92% under all tested conditions with lower variability.

Error metrics:

- MAE for PADC-MAC: percentages (11.75% high irradiance, 28.46% low irradiance).
- MAE for our model: Absolute value of 35 W/m^2 (about 8-12% depending on the range).

Reasons for better generalization in our model:

1) Visual properties carry useful information even in the absence of high-irradiance (cloud cover, ground state). Historical data becomes meaningless during prolonged periods of low irradiance.
2) Using a weighted loss function, more attention is paid to low-irradiance conditions (2$\times$ times more important if <30 W/m^2)
3) Low-irradiance conditions generated in the data-augmentation process.

2) EENA NARNET Analysis: Bhuvaneswari et al. [12] obtain 99.62% accuracy, numerically exceeding our best performance figures. However, direct comparisons are difficult due to differences in methodology used.

Limited evaluation scope:

- Tested only on "sunny" and "cloudy" days—no nighttime validation
- No range coverage evidence (0-1200 W/m^2 verification absent)
- Conference publication (lower peer review rigor)

Overfitting indicators:

- 99.62% accuracy is suspiciously high for real-world solar prediction
- No catastrophic failure analysis (our zero-failure hypothesis is proven explicitly)
- No performance degradation at varying irradiance levels (as demonstrated by our Table IV)

Reproducibility concerns:

- Insufficient information about methodology
- Lack of clear train/test split methodology may lead to data leakage issues
- Lack of validation with diverse weather patterns

Our evaluation focuses on coverage, robustness to edge cases, and practical performance metrics which allow us to evaluate the results not only by means of aggregated accuracy.

D. Advantages Summary: CNN vs Autoregressive

TABLE VI
DETAILED COMPARISON OF CNN VS AUTOREGRESSIVE APPROACHES

Aspect	NARNET	Our CNN
Input Type	Time-series	Time-series (2 features)
Feature Source	Temporal patterns	Temporal + derivative patterns
Accuracy Range	71-89%	**88-92%**
Consistency	Variable	**Stable**
Low-light	71-72%	**88-92%**
Error Metric	PER: 11.75-28.46%	MAE: **35 W/m^2**
Rapid transitions	Struggles	**Robust**
Feature engineering	Manual	**Automatic**
Edge cases	Degradation	**Zero fails**
Inference time	50ms	**15ms** (INT8)

E. Limitations and Future Work

1) Current Limitations:

1) **Imaging Device:**Use of an imaging sensor increases the cost and complexity of hardware integration.
2) **New Site Labeled Data:** For optimal performance, it is required to have labeled data for new deployment sites.
3) **Shorter Temporal Sequence Awareness:** The current implementation uses only the present moment to make predictions.
4) **Short-Horizon Prediction Focus:** At this point, the pipeline is limited to short-horizon forecasting.
5) **Nighttime Performance Limitation:** While nighttime performance is good, improvements are possible.

2) Future Enhancements:

1) **CNN-LSTM Combination:** Extend the current framework to include sequential awareness.
2) **Visual & Sensor Features Fusion:** Incorporate visual features and numerical sensor data (wind speed, temperature, etc.) via a multi-modality fusion algorithm:

$$\hat{I}(t) = f_{fusion}(CNN(V_t), MLP(S_t)) \tag{14}$$

where V_t denotes the visual modality and S_t denotes sensor features.

3) **Transfer Learning:** Perform pre-training at several sites followed by local fine-tuning with reduced data requirements
4) **Uncertainty Estimation:** See [13] to provide support for risk-aware control algorithms.

$$\text{Var}[\hat{I}] \approx \frac{1}{T}\sum_{i=1}^{T}(\hat{I}_i - \bar{\hat{I}})^2 \tag{15}$$

to support risk-aware control decisions.

5) **Federated Learning:** Facilitate collaborative learning without requiring centralized access to data.
6) **Attention Models:** Add spatial attention mechanisms to improve interpretability.
7) **Multi-Step Ahead Prediction:** Extend the output to $[\hat{I}(t+1), \hat{I}(t+2), \ldots, \hat{I}(t+H)]$ for longer-horizon scheduling.

F. Broader Impact

While our work concerns WSN operation specifically, the same formulation is equally applicable to a wide range of tasks where local renewable resource estimation is required for control, such as:

- **Smart grid solar forecasting:** Renewable integration into a distribution grid
- **Agricultural IoT:** Precision agriculture with adaptive irrigation
- **Building Energy Management System:**HVAC pre-cooling based on sunlight estimates

- **Electric Vehicle Charging Station:** Solar-powered charging station with power balance
- **Autonomous Drones:** Solar-powered autonomous vehicles with visual navigation capabilities
- **Space application:** Mars rovers and satellites capable of estimating local solar irradiance via image analysis

In general, the main discovery made in the paper is related to the fact that visual context may improve performance in weather-dependent prediction problems compared to temporal history alone.

VII. Conclusion

This paper describes a CNN-based architecture to predict irradiation levels suitable for use within solar power-enabled WSNs and examines the effectiveness of such an approach considering two criteria – predictive performance and operational readiness for deployment in edge environments. The best-performing model achieved the prediction accuracy rate of 88-92% in all the irradiation levels between 0 and 1200 W/m^2, obtaining the MAE of 35 W/m^2 and the mean bias of 6.12 W/m^2, and thus proving that machine learning on visual features leads to stable enough results for practical application within energy-conscious IoT devices.

The major take-away message from the research conducted was the idea that the model improvement process included multiple stages involving engineering tuning rather than any singular architectural change: the weighted loss led to better performance in low-lighting periods, while calibration eliminated the problem of systematic bias. Finally, the smoothing function helped solve the challenge with excessive variance at the transitions, which resulted in consistent prediction results in all three examined scenarios (nighttime, low irradiation, and high irradiation) without any errors, while still being applicable in edge devices due to low inference time (around 15 ms) after INT8 quantization, which required only slight loss of accuracy and insignificant energy consumption compared to harvested energy per day.

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